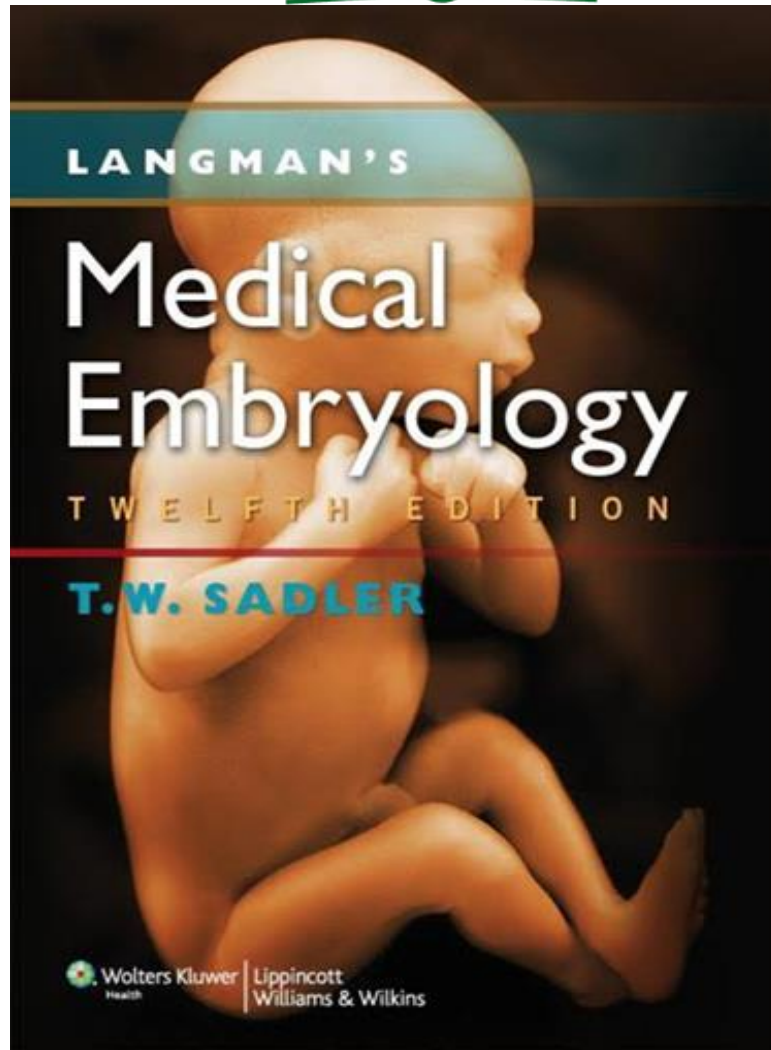




Alameed university-Faculty of medicine



General Embryology L1

Mitosis and Meiosis

Dr. Shehab Ahmad Faris

From a single cell to a baby in 9 months ,a developmental process that represents an amazing integration of increasingly complex phenomena. The study of these phenomena is called

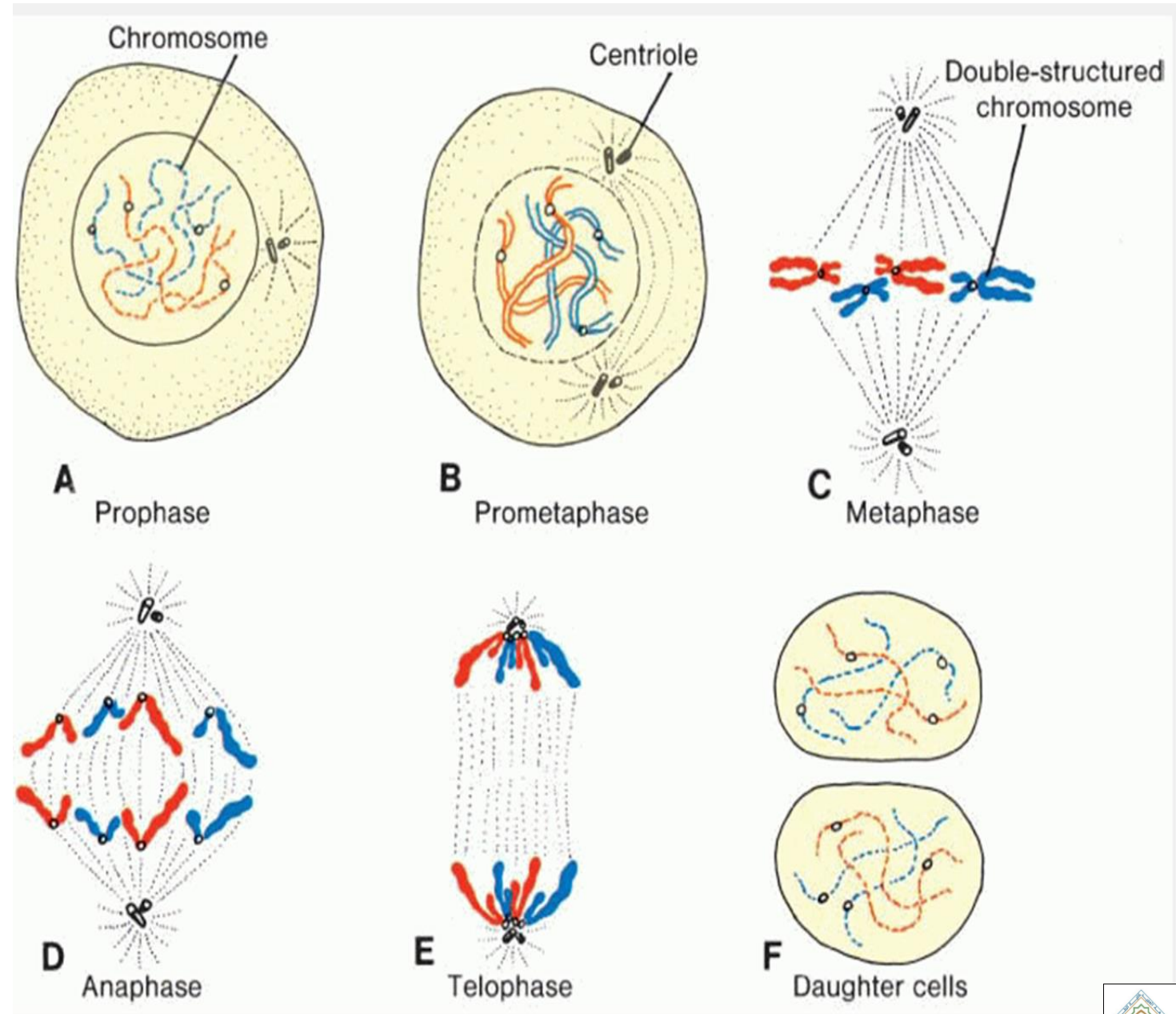
Embryology



Mitosis

-Is the process where by one cell divides, giving rise to 2 daughter cells that are genetically identical to the parent cell, each daughter cell contain the total 46 chromosomes.

-In somatic cells, chromosomes appear as 23 homologous pairs to form the diploid number of 46. There are 22 pairs of matching chromosomes, the autosomes, and one pair of sex chromosomes. If the sex pair is XX, the individual is genetically female; if the pair is XY, the individual is genetically male



Meiosis

-Meiosis is the cell division that takes place in the germ cells to generate male and female gametes, sperm and egg cells, respectively.

- Meiosis requires two cell divisions, **meiosis I** and **meiosis II**, to reduce the number of chromosomes to the haploid number of 23.

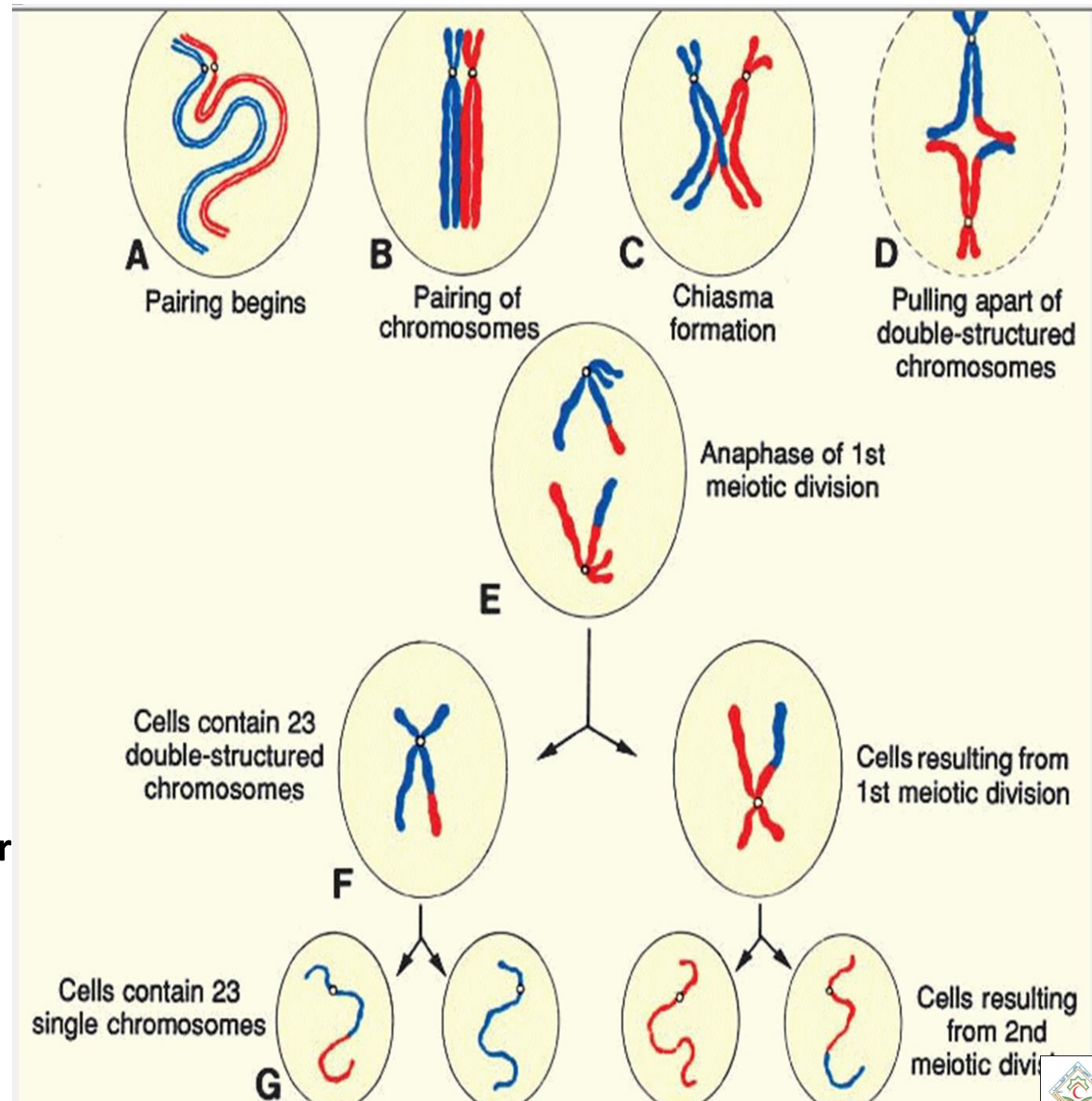
-As in mitosis, male and female germ cells (primary spermatocytes and primary oocytes) at the beginning of meiosis I replicate their DNA so that each of the 46 chromosomes is duplicated into sister chromatids.

-In contrast to mitosis, however, homologous chromosomes then align themselves in pairs, a process called **synapsis**. The pairing is exact and point for point except for the XY combination.

-Homologous pairs then separate into two daughter cells, thereby reducing the chromosome number from diploid to haploid.

-Shortly thereafter, meiosis II separates sister chromatids.

-Each gamete then contains 23 chromosomes



Crossover

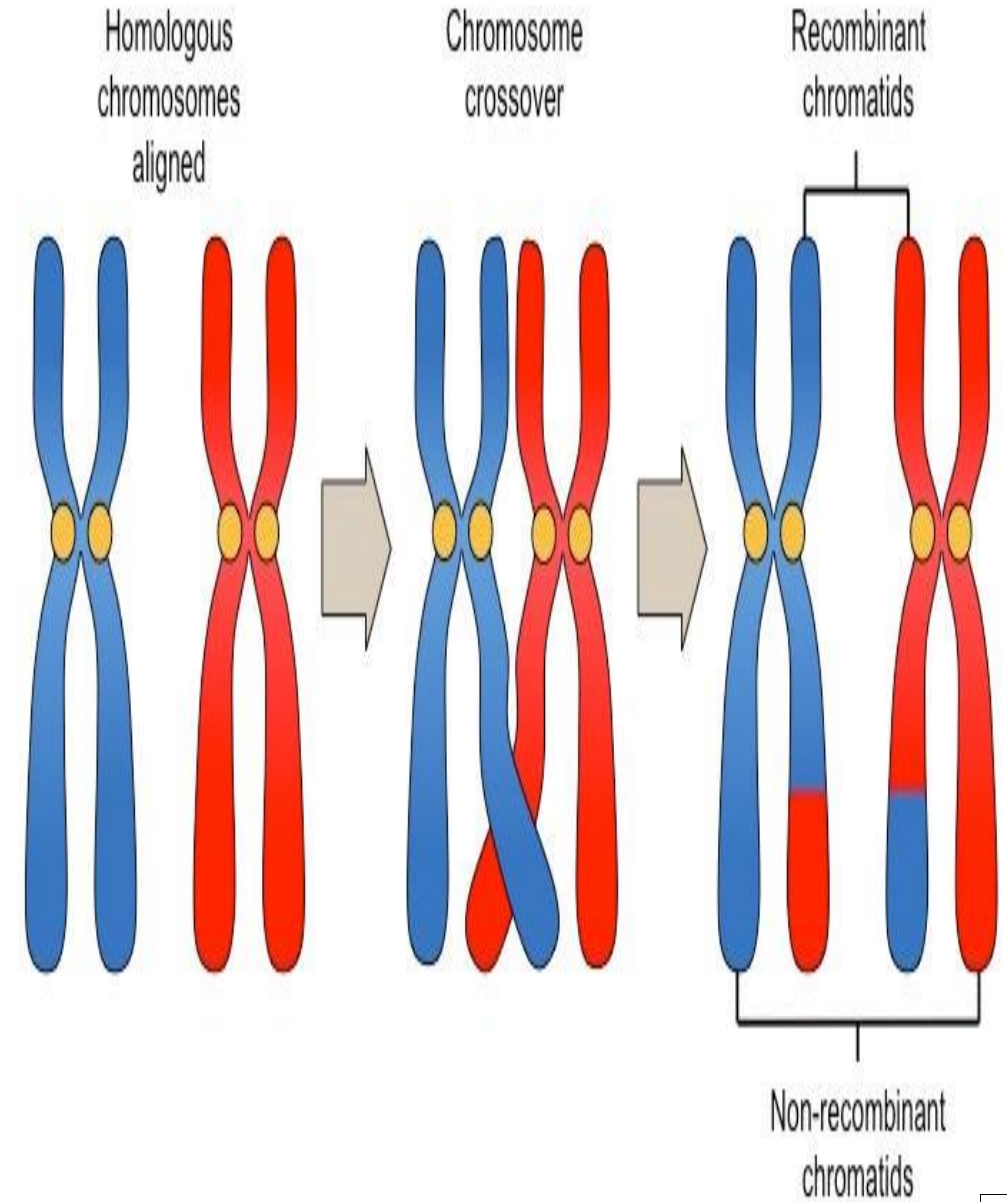
1-Critical event in meiosis 1

2-Is the interchange of chromatid segments between paired homologous chromosomes

3-It occurs during separation

Chiasma: points of interchange are temporarily united and form an X-like structure

4- (30-40) crossovers(1-2 per chr.) with each meiotic 1 division are most frequent between genes that are far apart on chromosome



Results of meiotic divisions

1-Genetic variability is enhanced through

a. Crossover, which redistributes genetic material

b. Random distribution of homologous chromosomes to the daughter cell

2-Each germ cell contains a haploid number of chromosomes so that(at fertilization) the diploid number of 46 is restored



Chromosomal abnormalities:

1-Numerical (nondisjunction)

2-Translocations

3-Structural

4-Fragile site

-It is estimated that 50% of conceptions end in spontaneous abortion

-50% of these due to chromosomal abnormalities.

-So 25% of conceptuses have a major defects

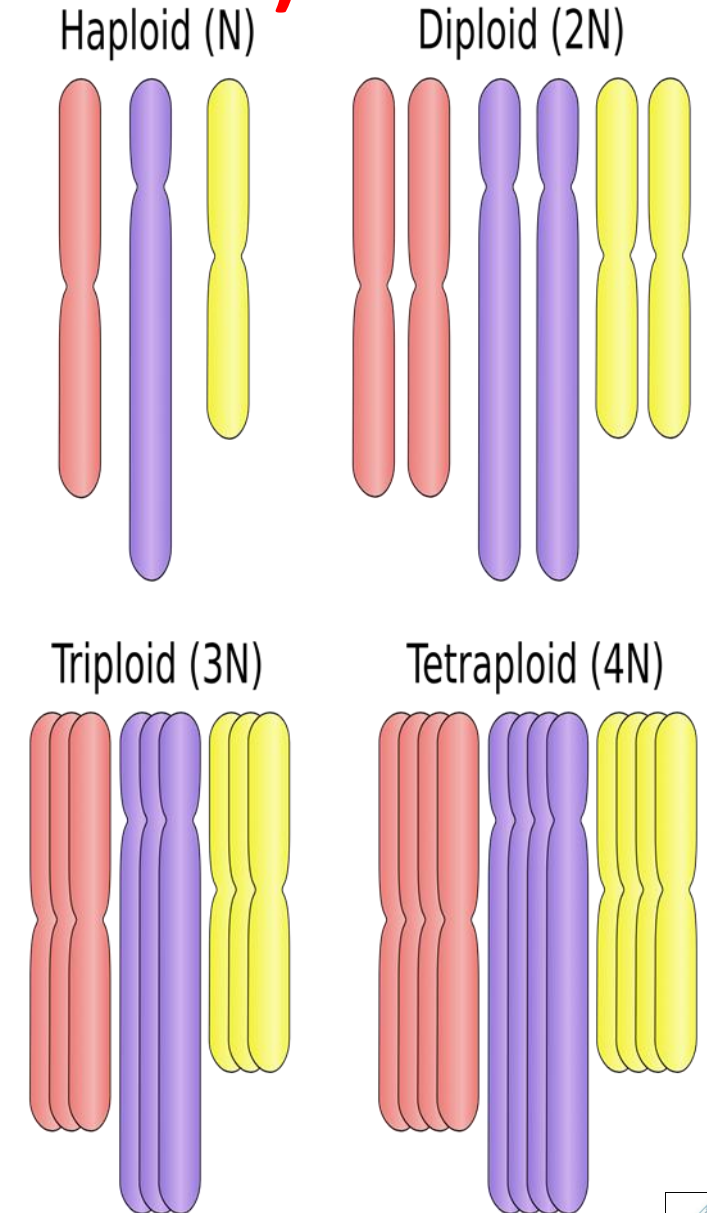
-Chromosomal abnormalities account for 10% major birth defects

-Gene mutation account for additional 8% major birth defects



1. Numerical abnormalities (nondisjunction)

- Normal human somatic cell contain 46 chromosomes (**diploid**) or $2n$
- Normal gamete contain **23**.(**haploid**) or n
- Euploid**: Refers to any exact multiple of n =diploid or triploid.
- Aneuploid**: Refers to any chromosomes number that is not euploid
- Trisomy= **extra** chromosome **is present**
- Monosomy= **1** chromosome **is missing**
- Abnormalities in chromosome number may originate during meiotic or mitotic divisions

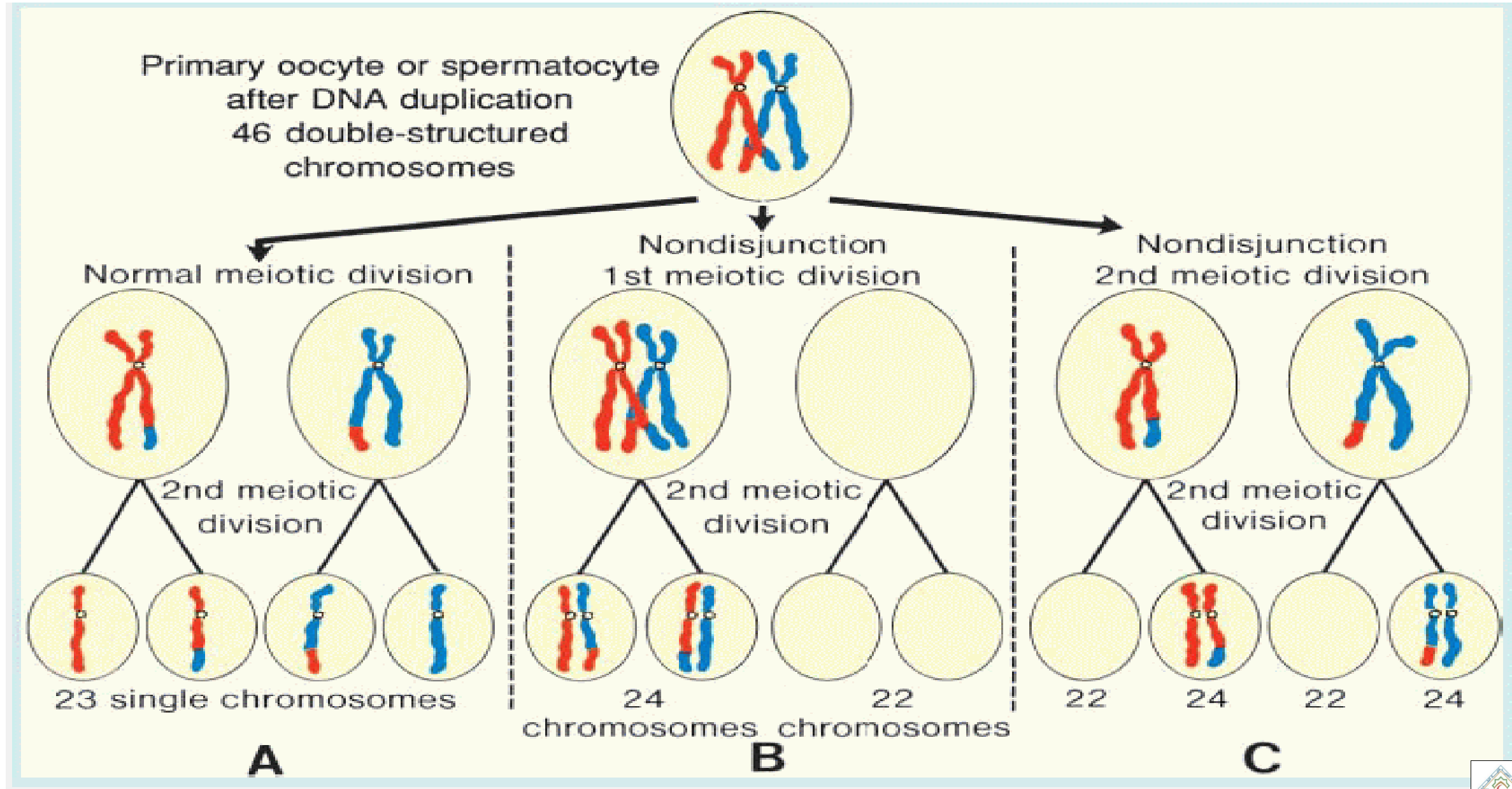


Nondisjunction

meiosis	mitosis
Commonly	occasionally
(Nondisjunction) failure of separation of the paired chr. during meiotic division.	(mitotic nondisjunction) =mosacisim i.e. some cells have normal & others are abnormal.
Occur either in 1 st or 2 nd meiotic division	in the embryonic cell during the earliest cell divisions
As result one cell receive 22 chr, & the other 24 chr.	. some cells have normal numbers of chr.23 & others have abnormal numbers
In women the incidence increase with age especially at 35 yrs & older	same
All have characteristics Features known as syndromes	Affected individuals may exhibit few or many of the characteristics of a particular syndromes depending on the number of cells involved and their distribution.



Meiotic nondisjunction



2. Traslocations

Some times chromosomes break, & pieces of one chromosome attach to another.

2 types:

1-Balanced:

beakage & reunion occur between 2 chromosomes, but no critical genetic material is lost & individuals are normal.

2-Unbalanced:

In which part of the chromosome is lost & an altered phenotype is produced.

E.g. unbalanced translocation between the long arm of chromosomes 14 & 21 during meiosis 1 or 2 produce gametes with an extra copy of chromosome 21 ,one of the causes of Down syndrome



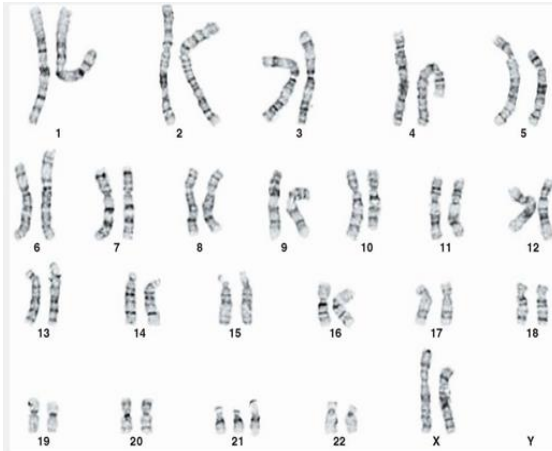
Trisomy 21 (down syndrome)

Causes:

- 95% caused by an extra copy of chr 21(trisomy21)
- 4% due to unbalanced translocation
- 1% is caused by mosaicism resulting from mitotic nondisjunction

Features include:

- Varying degrees of mental retardation,cardiac defects and hypotonia
- Craniofacial abnormalities include
 - 1-upward slanting eyes
 - 2- flat facies
 - 3-small ears
- They have high incidence of leukemia,infections,thyroid dysfunction. Premature aging
- Nearly all develop signs of Alzheimer's disease after age 35
- Incidence is 1 in 2000 conceptuses for women under age 25, 1 in 300 at age 35, 1 in 100 at age 40



DOWN SYNDROME PHYSICAL CHARACTERISTICS



Klinefilter syndrome

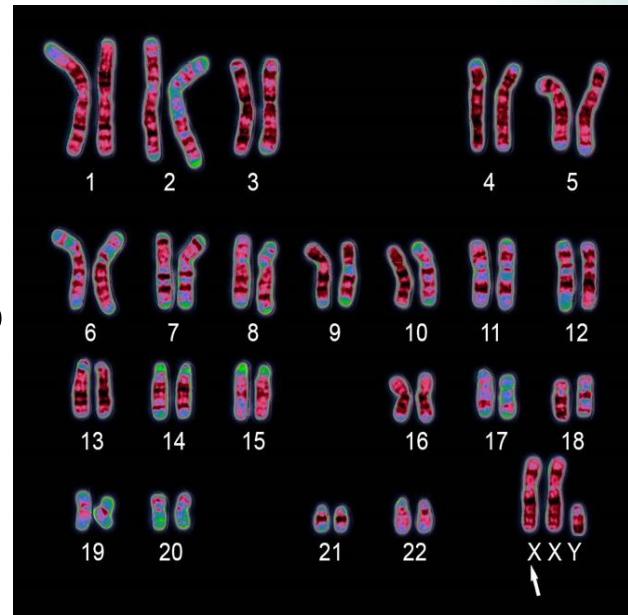
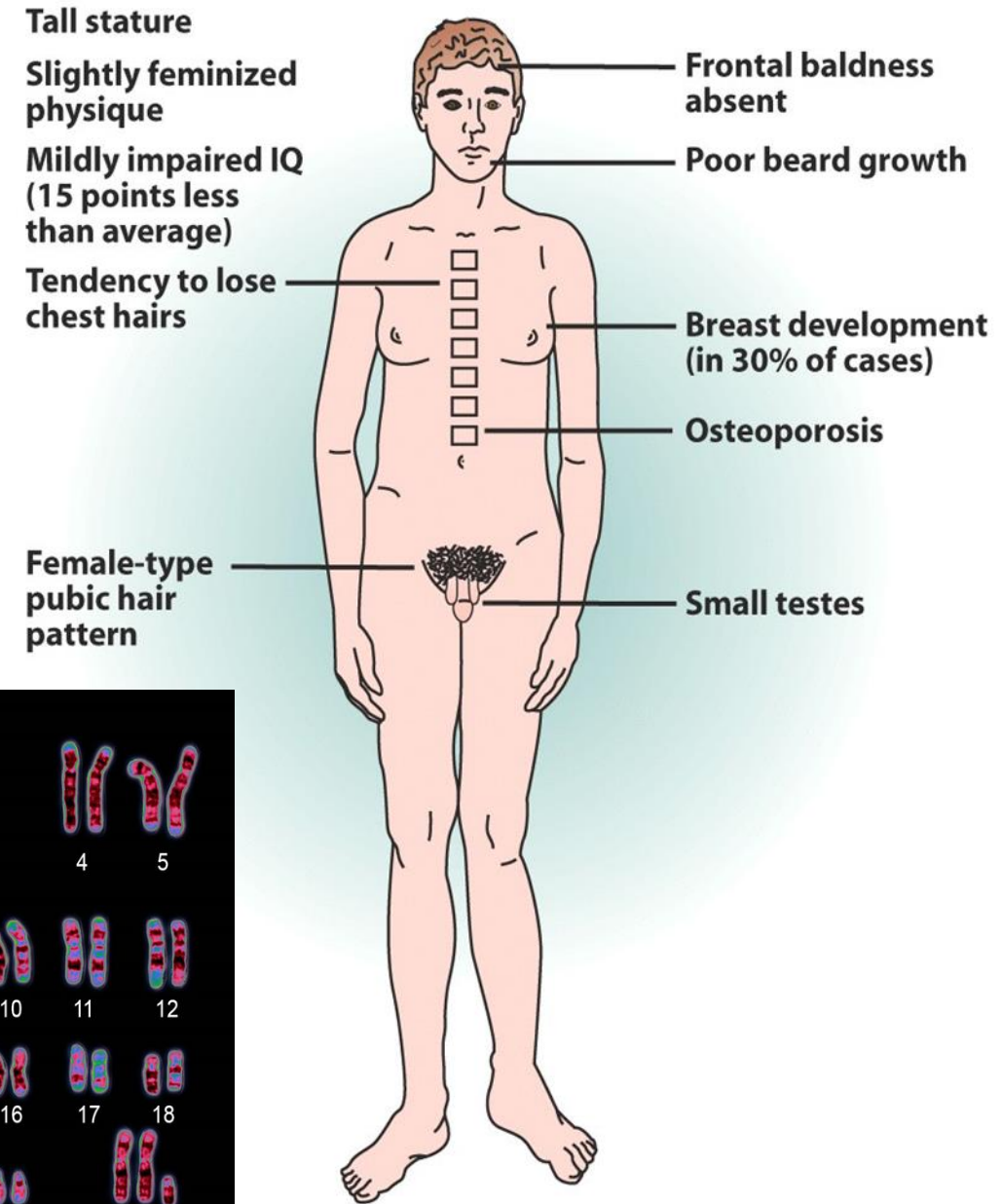
Causes:

- Nondisjunction of the XX homologues is the most common causative event.
- The cells have 47 chromosome & with a sex chromosomal complement of the XXY type

Features include:

sterility, testicular atrophy, hyalinization of the seminiferous tubules, and usually gynecomastia found only in males and usually detected at puberty,

- Sex chromatin body is found in 80% of cases
- The incidence is about 1 in 500 males



Turner syndrome

Causes:

55% of these women are monosomic for x chromosome & chromatin negative due to meiotic nondisjunction

In the remainder it is either due to

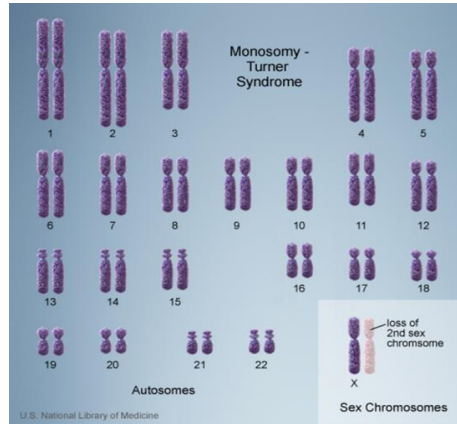
1- structural abnormalities of the x chromosome

2-mitotic nondisjunction resulting in mosaicism

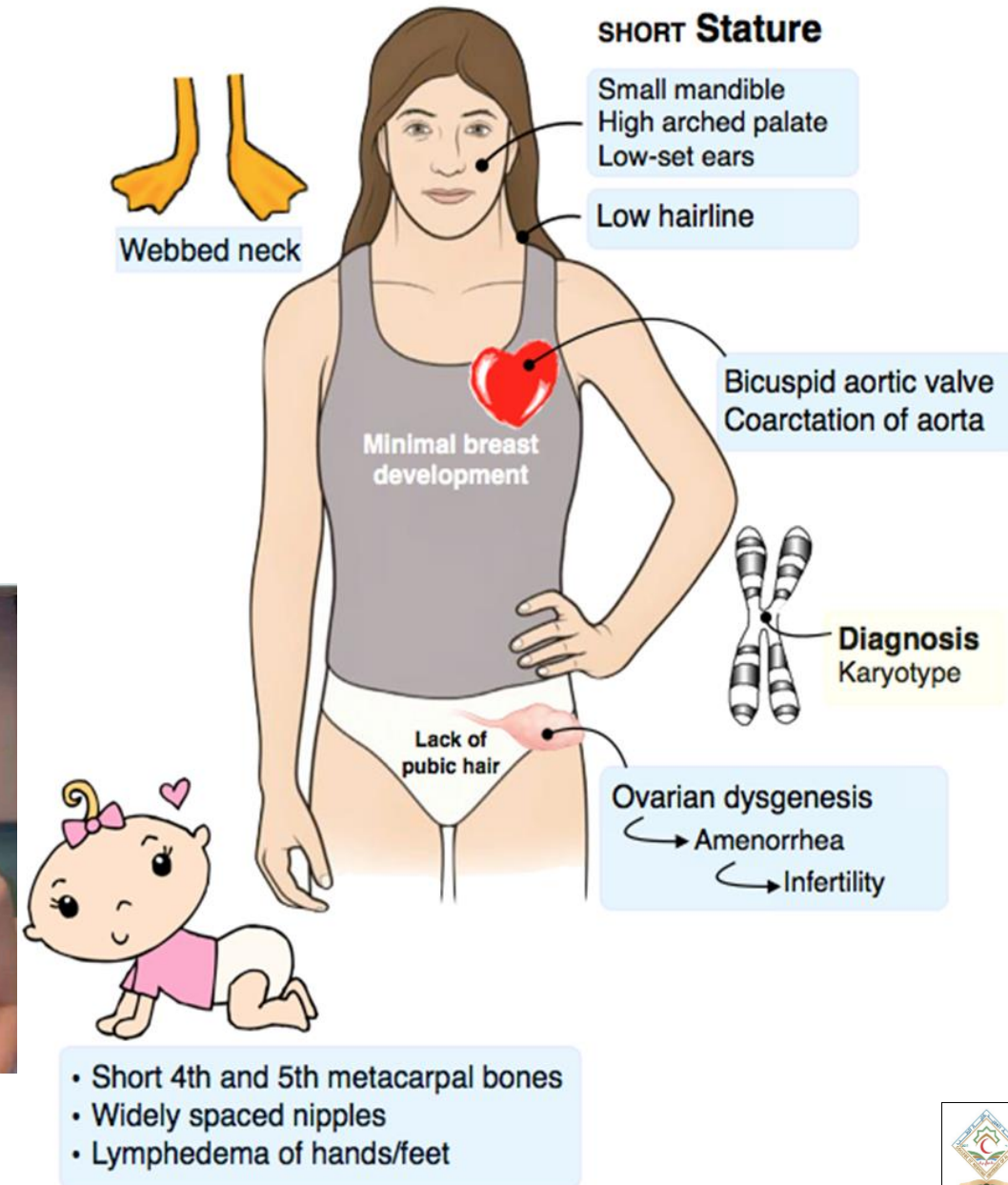
is the only monosomy compatible with life. Even then, 98% of all fetuses with the syndrome are spontaneously aborted. The few that survive are unmistakably female in appearance

Features include:

- Gonadal dysgenesis
- Short stature
- Web neck
- Lymphedema of extremities
- Skeletal deformities
- Broad chest & widely spaced nipples



Turner Syndrome 45 XO



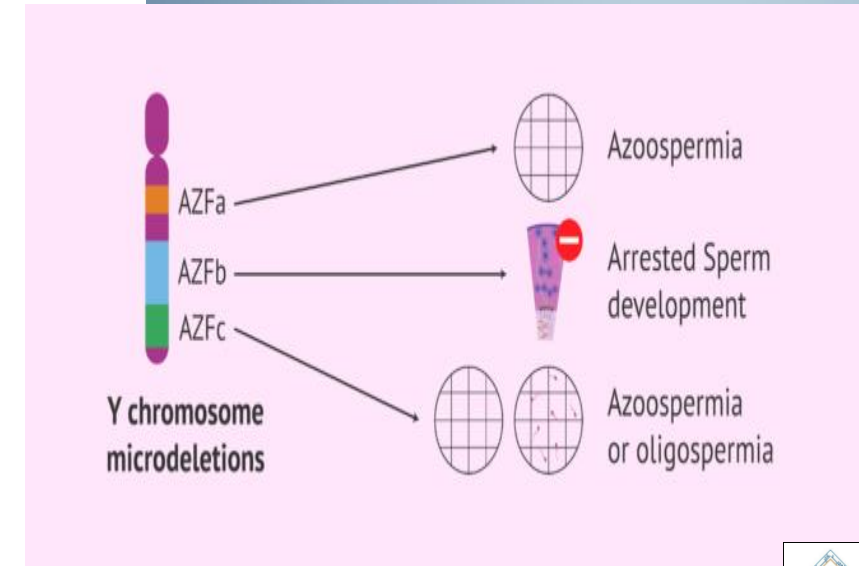
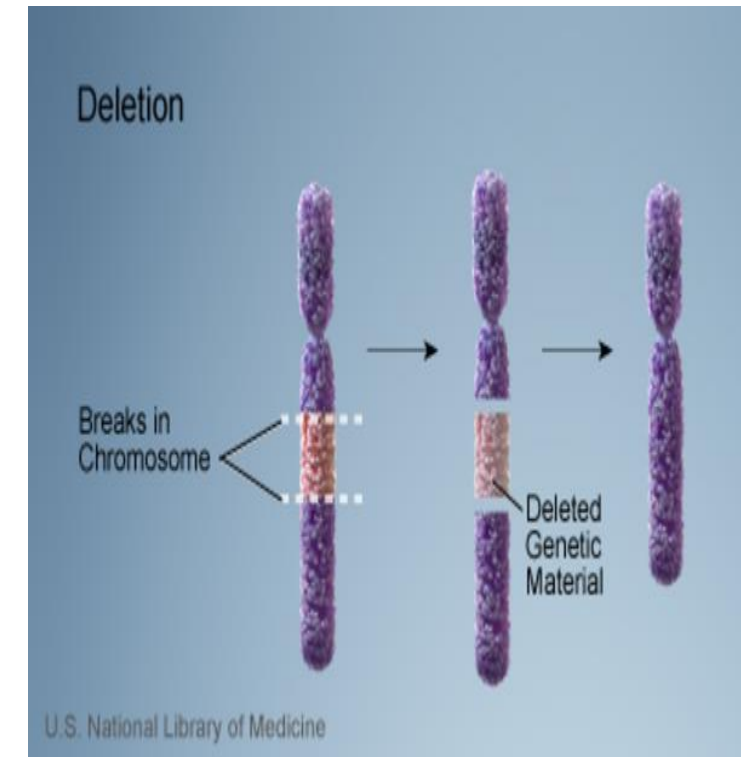
3. Structural Abnormalities

A. Structural chromosome abnormalities

result from chromosome breakage. Breaks are caused by environmental factors, such as **viruses, radiation, and drugs**. The infant with partial deletion of a chromosome is abnormal, when the broken piece of a chromosome is lost

B. Microdeletions

spanning only a few contiguous genes, may result in microdeletion syndrome or contiguous gene syndrome. Sites where these deletions occur, called contiguous gene complexes, can be identified by high-resolution chromosome banding.



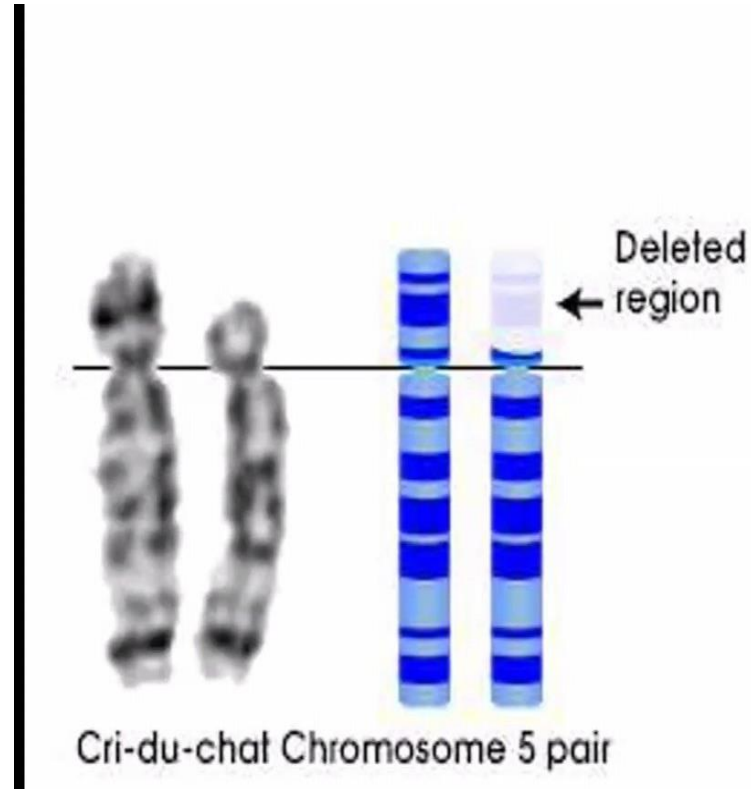
cri-du-chat syndrome

Causes:

partial deletion of the short arm of chromosome 5

Features

- 1.a cat-like cry
- 2.microcephaly
- 3.mental retardation
- 4.congenital heart disease



Angelman syndrome

Causes:

Microdeletion occur on the long arm of chromosome 15

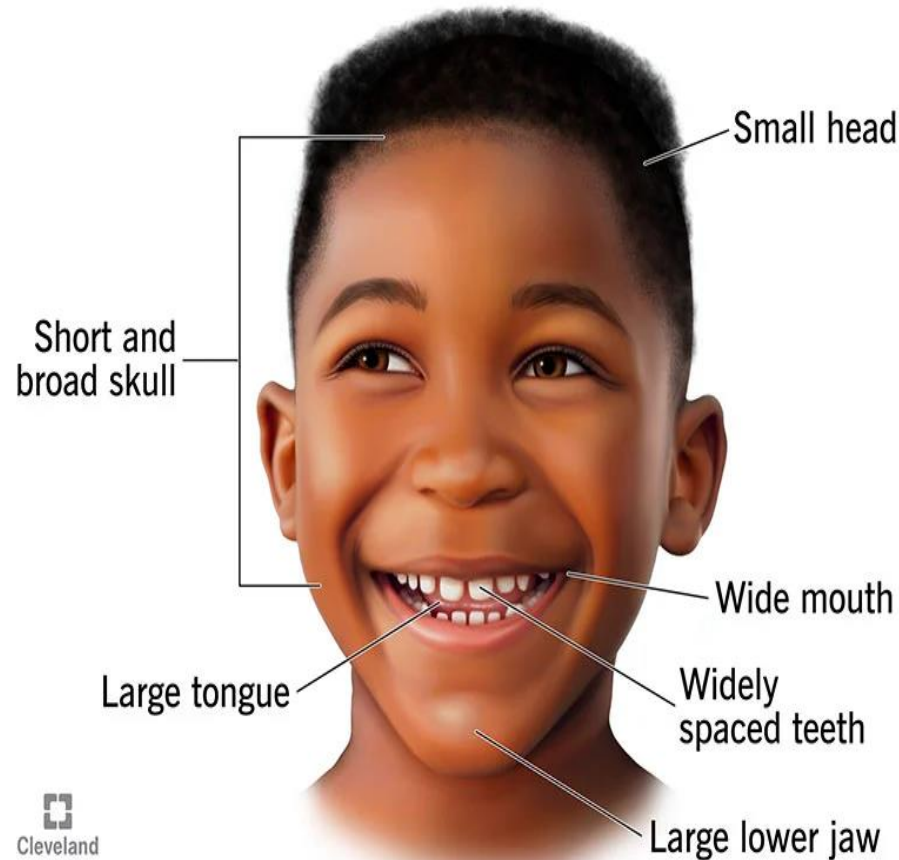
Inheriting the deletion on the **maternal chromosome** results in **Angelman syndrome**.

The children have

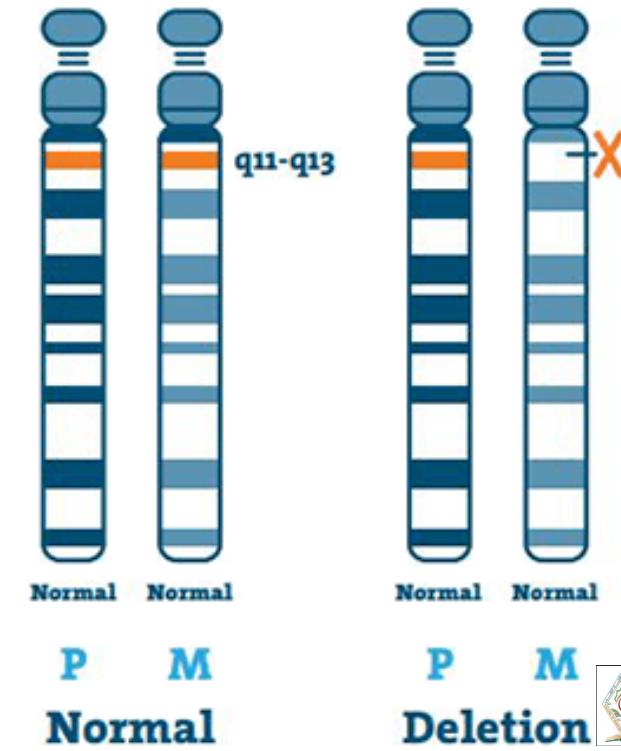
Features include:

- Mental retardation
- Cannot speak
- Poor motor development
- Prone to unprovoked & prolonged periods of laughter

Angelman syndrome



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Prader-willi syndrome

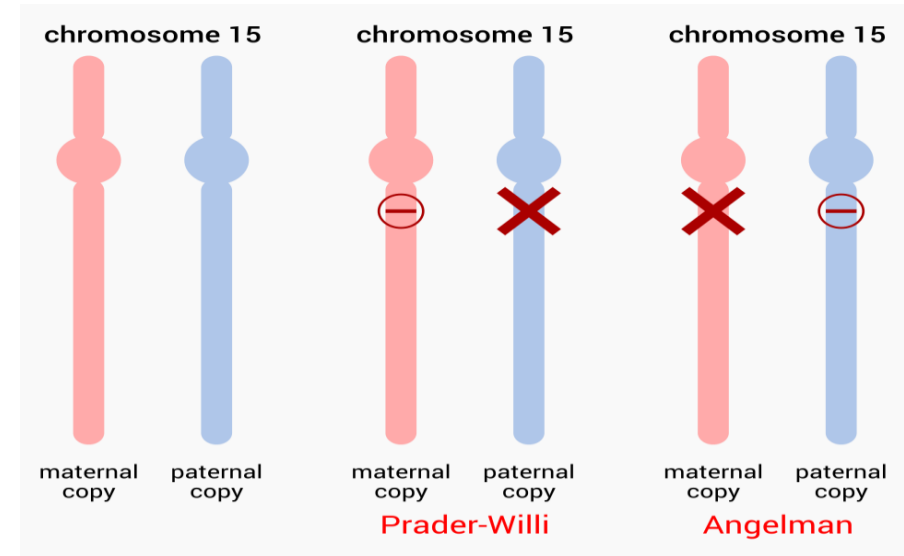
Causes:

If the defect (microdeletion on the long arm of chromosome 15) inherited on the paternal chromosome

Prader-willi syndrome is produced

Features include:

- Hypotonia
- Obesity
- Mental retardation
- Hypogonadism
- Cryptorchidism



4. Fragile site

Regions of the chromosome that demonstrate a propensity to separate or break under certain cell manipulations.

E.g. Fragile x chromosome

Features include:

1-Mental retardation

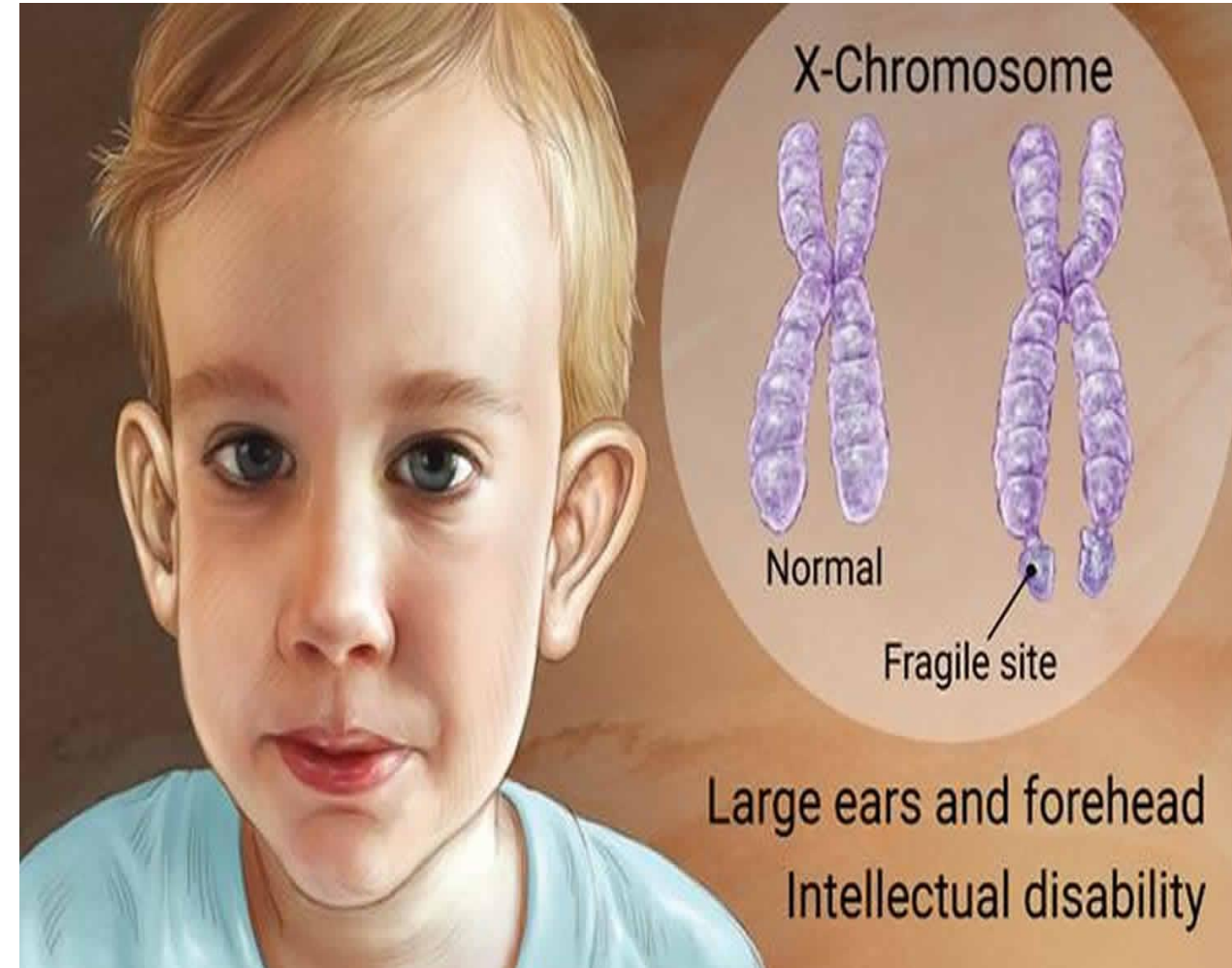
2-Large ears

3-Prominent jaw

4-Pale blue iris

-Male affected more than female

-2nd only to down syndrome as a cause of chromosomal **mental retardation**



Gene mutation

change in the structure or function of single gene so called single gene mutation

Equal to 8% of all human malformation

Dominant mutation : single allele is affected

Recessive mutation : double alleles are affected

Or the mutation is x-linked

All chromosomes have double alleles(dose of genetic determinants)

Except x, y chromosomes

Achondroplasia

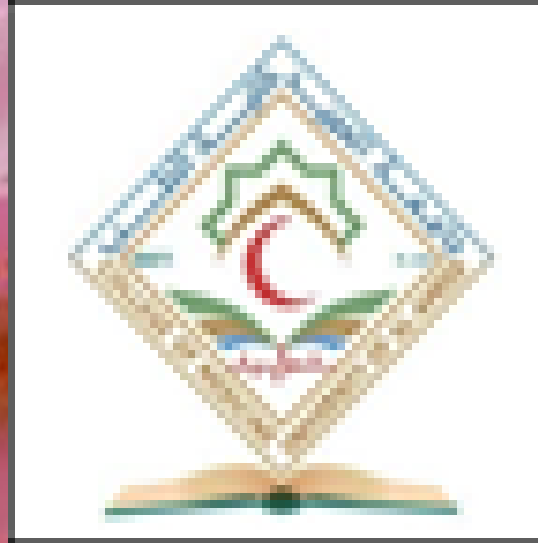
- Caused by a gene mutation

- Shown to be associated with advanced paternal age.

- Gene mutation affects bone formation



Thank
You



د. شہاب احمد قاری

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